

Question 1.

Let X be a random variable with probability distribution function:

$$f(x) = \begin{cases} 0.2, & \text{for } |x| < 1 \\ 0.1, & \text{for } 1 < |x| < 4 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(0.5 < x < 5)$ is -----.

Question 2.

Write the sample space S for the following random experiments:

- (a) We toss a coin until we see two consecutive tails. We record the total number of coin tosses.
- (b) A bag contains 44 balls: one red, one blue, one white, and one green. We choose two distinct balls and record their color in order.
- (c) A customer arrives at a bank and waits in line. We observe T , which is the total time (in hours) the customer waits in line. The bank has a strict policy that no customer waits more than 20 minutes under any circumstances.

Question 3.

Let A , B , and C be three events in the sample space S . Suppose we know $A \cup B \cup C = S$, $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$, and $P(A \cup B) = \frac{5}{6}$. Then find the following:

- (a) $P(A \cap B)$
- (b) Do A , B , and C form a partition of S ?
- (c) $P(C - (A \cup B))$
- (d) If $P(C \cap (A \cup B)) = \frac{5}{12}$, find $P(C)$

Question 4.

A box of cartridges contains 30 cartridges, of which 6 are defective. If 3 of the cartridges are removed from the box in succession without replacement, what is the probability that all 3 cartridges are defective?

Question 5.

Let A and B be two events such that $P(A^C) = 0.4$ and $P(A \cap B) = 0.2$. Then $P(A \cap B^C)$ is equal to -----.

Question 6.

Let A and B be two events such that $P(A) = \frac{1}{5}$ and $P(A \text{ or } B) = \frac{1}{2}$. Let $P(B) = P$. For what values of P are A and B independent?

Question 7.

Find the expectation of a random variable X if

$$f(x) = \begin{cases} ke^{-x} & \text{for } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Question 8.

Consider two random variables X and Y with joint PMF given in the table:

	$Y = 1$	$Y = 2$	$Y = 3$
$X = 1$	1/12	1/24	1/24
$X = 2$	1/6	1/12	1/8
$X = 3$	1/4	1/8	1/12

- (a) Find $P(X \leq 2, Y \leq 4)$
- (b) Find the marginal PMFs of X and Y
- (c) Find $P(Y = 2|X = 1)$
- (d) Are X and Y independent?

Question 9.

Let X be a random variable with PDF given by

$$f_X(x) = \begin{cases} cx^2, & |x| \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the constant c
- (b) Find $E(X)$ and $Var(X)$
- (c) Find $P(X \geq \frac{1}{2})$

Question 10.

A die is loaded in such a way that the probability of the face with j dots turning up is proportional to j for $j = 1, 2, \dots, 6$. What is the probability, in one roll of the die, that an odd number of dots will turn up?

Question 11.

Two boxes containing marbles are placed on a table. The boxes are labeled B1 and B2. Box B1 contains 7 green marbles and 4 white marbles. Box B2 contains 3 green marbles and 10 yellow marbles. The boxes are arranged so that the probability of selecting box B1 is $1/3$ and the probability of selecting box B2 is $2/3$. Kathy is blindfolded and asked to select a marble. She will win a color TV if she selects a green marble.

1. What is the probability that Kathy will win the TV (that is, she will select a green marble)?
2. If Kathy wins the color TV, what is the probability that the green marble was selected from the first box?

Question 12.

One-half percent of the population has AIDS. There is a test to detect AIDS. A positive test result is supposed to mean that you have AIDS but the test is not perfect. For people with AIDS, the test misses the diagnosis 2% of the times. And for the people without AIDS, the test incorrectly tells 3% of them that they have AIDS.

1. What is the probability that a person picked at random will test positive?
2. What is the probability that you have AIDS given that your test comes back positive?

Question 13.

Two points are taken at random on the given straight line of length a . Prove that the probability of their distance exceeding a given length c (where a is greater than c) is equal to $\left(1 - \frac{c}{a}\right)^2$.

Question 14.

Is the real-valued function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 + |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

a probability density function for some random variable X?

Question 15.

What is the probability density function of the random variable whose cdf is

$$F(x) = \frac{1}{1 + e^{-x}}, -\infty < x < \infty \text{ ?}$$

Question 16.

What is the 87.5 percentile for the distribution with density function

$$f(x) = \frac{1}{2}e^{-|x|} - \infty < x < \infty \text{ ?}$$

Question 17.

Let X and Y be discrete random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{x+y}{32} & \text{for } x = 1, 2; y = 1, 2, 3, 4 \\ 0 & \text{otherwise.} \end{cases}$$

What is the conditional probability of Y given X = x ?

Question 18.

Let X and Y be two independent random variables with identical probability density function given by

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

What is the probability density function of $W = \min\{X, Y\}$

Question 19.

Let X and Y have joint density function

$$f(x, y) = \begin{cases} 2e^{-x-y} & \text{for } 0 < x \leq y < \infty \\ 0 & \text{otherwise.} \end{cases}$$

What is the marginal density of X where nonzero?

Question 20.

Let X be a random variable with

$$E(X^n) = 0.8 \text{ for } n = 1, 2, 3, \dots, \infty.$$

What are the moment generating function and probability density function of X ?

Question 21.

If the moment generating function of a random variable X is given by

$$M(t) = \frac{5}{15}e^t + \frac{4}{15}e^{2t} + \frac{3}{15}e^{3t} + \frac{2}{15}e^{4t} + \frac{1}{15}e^{5t},$$

then what is the probability density function of X ? What is the space of the random variable X ?

Question 22.

What are the mean and variance of a distribution whose moment-generating function is $m_X(t) = e^{4(e^t - 1)}$?

Question 23.

If the joint moment generating function of the random variables X and Y is

$$M(s, t) = e^{(s+3t+2s^2+18t^2+12st)}$$

what is the covariance of X and Y ?

Question 24.

Let X be a random variable with PDF $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, show that

$$\phi_X(w) = \frac{\lambda}{\lambda - iw}$$